

Model, language inputs, and references

Ideal double-integrator dynamics

$$\dot{x} = (p^T, v^T)^T, \quad \dot{x} = Ax + Bu, \quad A = \begin{bmatrix} 0 & I \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ I \end{bmatrix}$$

First natural-language input: goal selection

L_1 : = goal selection language

Second natural-language input: HRRT-star route selection

L_2 : = HRRT– star route selection language

Expert reference selected using both language inputs

$$\dot{x}_d(E, L_1, L_2, t) = Ax_d(E, L_1, L_2, t) + Bu_d(E, L_1, L_2, t)$$

LLM reference predicted using the goal language only

$$\dot{\hat{x}}_d(\hat{E}, L_1, t) = A\hat{x}_d(\hat{E}, L_1, t) + B\hat{u}_d(\hat{E}, L_1, t)$$

Ideal double integrator: $\zeta = \hat{\zeta} = 0$. Only planner and perception mismatch remain.

Controllers and exact deviation

Available LLM controller

$$u_{\text{VLM}}(x, \hat{E}, L_1, t) = \hat{u}_d(\hat{E}, L_1, t) - K[x - \hat{x}_d(\hat{E}, L_1, t)]$$

Nominal actual-scene expert controller

$$u_{c,d}(x, E, L_1, L_2, t) = u_d(E, L_1, L_2, t) - K[x - x_d(E, L_1, L_2, t)]$$

Exact LLM-to-expert control deviation

$$\Delta u_{\text{VLM}} = \hat{u}_d(\hat{E}, L_1, t) - u_d(E, L_1, L_2, t) + K[\hat{x}_d(\hat{E}, L_1, t) - x_d(E, L_1, L_2, t)]$$

Ideal double integrator: $\zeta = \hat{\zeta} = 0$. Only planner and perception mismatch remain.

Separate planner and perception mismatch

Planner reference mismatch on the perceived environment

$$d_x^{\text{plan}} := \hat{x}_d(\hat{E}, L_1, t) - x_d(\hat{E}, L_1, L_2, t)$$

Planner input mismatch on the perceived environment

$$d_u^{\text{plan}} := \hat{u}_d(\hat{E}, L_1, t) - u_d(\hat{E}, L_1, L_2, t)$$

Perception-induced reference mismatch

$$d_x^{\text{env}} := x_d(\hat{E}, L_1, L_2, t) - x_d(E, L_1, L_2, t)$$

Perception-induced input mismatch

$$d_u^{\text{env}} := u_d(\hat{E}, L_1, L_2, t) - u_d(E, L_1, L_2, t)$$

Total gaps

$$d_x = d_x^{\text{plan}} + d_x^{\text{env}}, \quad d_u = d_u^{\text{plan}} + d_u^{\text{env}}$$

Ideal double integrator: $\zeta = \hat{\zeta} = 0$. Only planner and perception mismatch remain.

Single residual and conformity score

Deviation split by uncertainty source

$$\Delta u_{\text{VLM}} = (d_u^{\text{plan}} + Kd_x^{\text{plan}}) + (d_u^{\text{env}} + Kd_x^{\text{env}})$$

Tracking-error dynamics

$$e := x - x_d(E, L_1, L_2, t), \quad A_c := A - BK, \quad \dot{e} = A_c e + R_\Sigma$$

Complete residual

$$R_\Sigma = B(d_u + Kd_x) = B(d_u^{\text{plan}} + Kd_x^{\text{plan}}) + B(d_u^{\text{env}} + Kd_x^{\text{env}})$$

Equivalent direct form used for calibration

$$R_\Sigma = B[\hat{u}_d(\hat{E}, L_1, t) - u_d(E, L_1, L_2, t) + K(\hat{x}_d(\hat{E}, L_1, t) - x_d(E, L_1, L_2, t))]$$

One trajectory-level conformity score

$$S_j^P = \sup_{t \in [0, T_j]} \|P^{1/2} R_{\Sigma, j}(t)\|_2$$

Ideal double integrator: $\zeta = \hat{\zeta} = 0$. Only planner and perception mismatch remain.

Lyapunov contraction and tracking tube

Constant-metric contraction condition

$$A_c^\top P + PA_c \leqslant -2\alpha_P P$$

Metric tracking error

$$W(t) := \|P^{1/2}[x(t) - x_d(E, L_1, L_2, t)]\|_2$$

Lyapunov inequality

$$D^+ W \leqslant -\alpha_P W + \|P^{1/2} R_\Sigma\|_2$$

Covered-episode implication

$$S_j^P \leq q_j^P \implies W(t) \leq W(0)e^{-\alpha_P t} + \frac{q_j^P}{\alpha_P}(1 - e^{-\alpha_P t})$$

Euclidean expert-relative tracking tube

$$\|x(t) - x_d(E, L_1, L_2, t)\|_2 \leq \frac{W(0)e^{-\alpha_P t} + \frac{q_j^P}{\alpha_P}(1 - e^{-\alpha_P t})}{\sqrt{\lambda_{\min}(P)}}$$

Ideal double integrator: $\zeta = \hat{\zeta} = 0$. Only planner and perception mismatch remain.